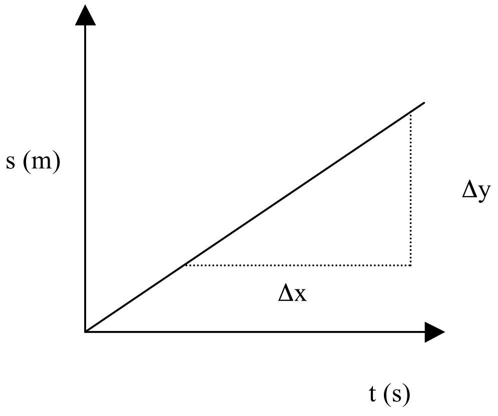
# GRAPHING MOTION

## Displacement - time

- ***Constant Velocity***



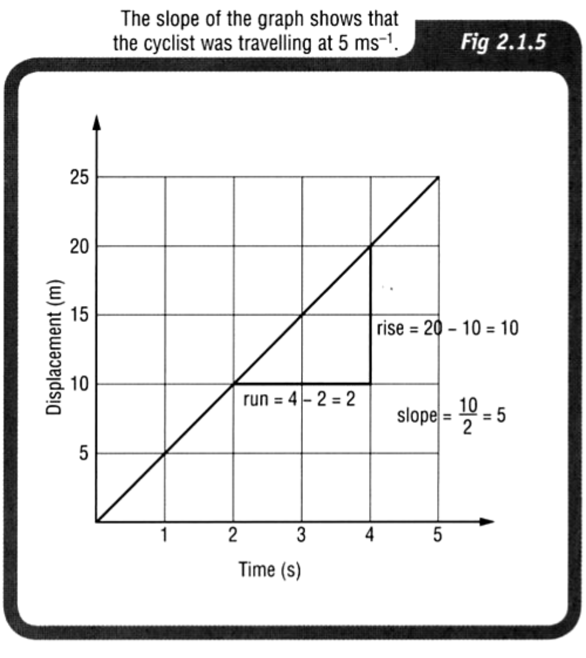

The slope or gradient of the line is given by:

$$\boxed{\text{slope} = \frac{\Delta y}{\Delta x} = \frac{s}{t}}$$

Hence the slope of the graph gives the ***constant velocity*** of the body or object.

## Examples

1. A cyclist moving at a constant velocity of 5.00 $ms^{-1}$ would generate the following displacement - time graph. It would have a constant gradient as shown below.




In maths you have probably learnt that:

$$\boxed{\text{gradient (slope)} = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x}}$$

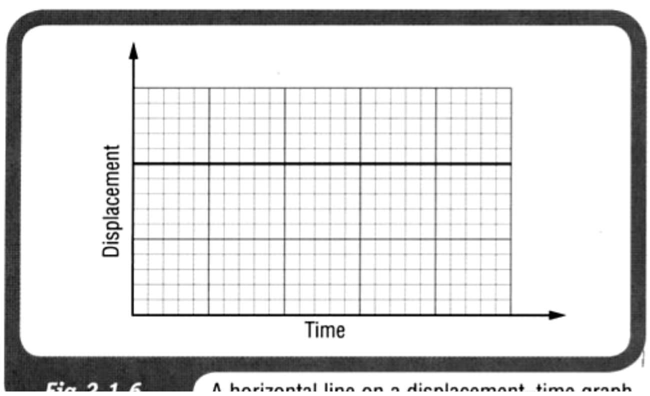
For this graph:

$$\text{gradient (slope)} = \frac{(10 - 0)}{(2 - 0)}$$

$$= 5.0 \text{ ms}^{-1}$$

2. A displacement–time graph like that shown indicates a vehicle that is stationary (not moving). This is because the rise is zero, so the slope is zero.

   Therefore the ***velocity is zero***.



The greater the slope of displacement time graph therefore, the greater the velocity, as shown in the graph below.

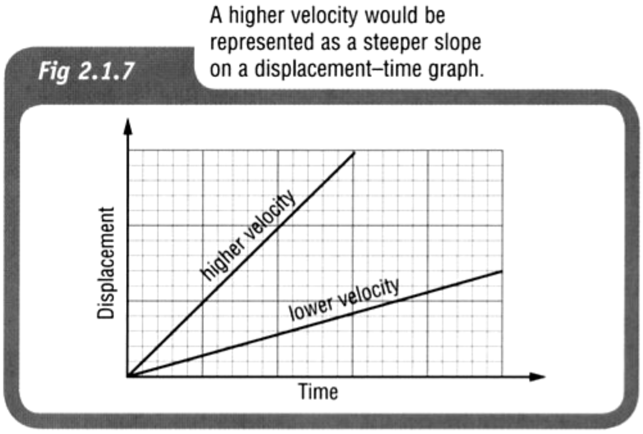
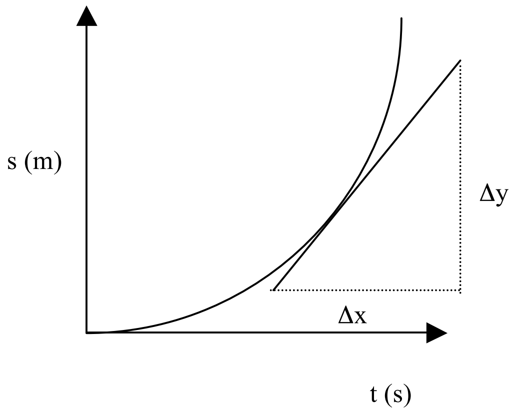


Fig 2.1.7 A higher velocity would be represented as a steeper slope on a displacement–time graph.

- ***Constant Acceleration***




The graph at left represents an object falling under gravity or one that is uniformly accelerated across the ground.

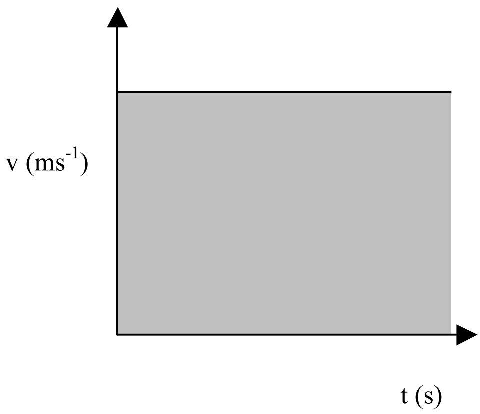
The slope at a point on the curve gives the ***instantaneous velocity*** of the object at that time.

The graph is a parabola and clearly suggests that $s\ \mu\ t^2$.

i.e. $s = ut + {}^{1}/_{2}\, at^2$

**Velocity - time**

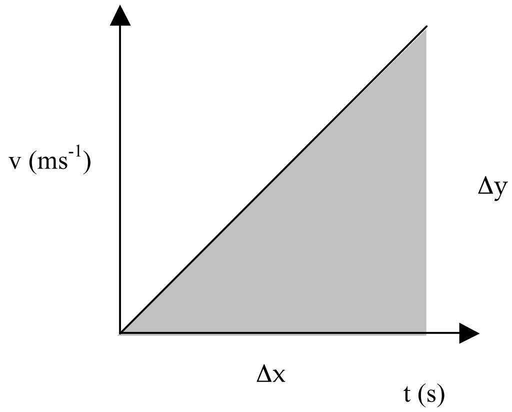
- ***Constant Velocity***




***The area under a v - t graph represents the displacement for the body.***

i.e. $s = v \times t$

- ***Constant Acceleration***




The slope gives the ***acceleration*** of the body.

$$\text{slope} = \frac{\Delta v}{\Delta t} = a$$

Area under graph = displacement

= 1/2 x base x height

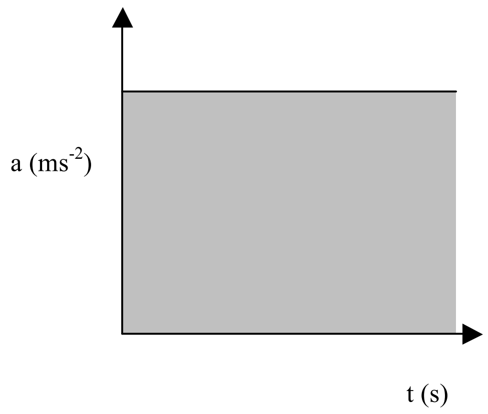
**Example**

A car accelerates uniformly from a stop sign and reaches 15.0 $ms^{-1}$ after 10.0 s. It maintains this velocity for 15.0 s before decelerating uniformly to a stop in 5.00 s at a set of traffic lights. The entire motion occurs in a straight line.

(a) Draw a velocity - time graph for the motion.
(b) Calculate the displacement of the car in this time.

**Acceleration - Time**

- ***Constant Acceleration***




***Area under graph = change in velocity***
***= $\Delta v$***

**Example**

Draw the acceleration - time graph for the car motion in the previous example.